Anomaly Detection in Browser Fingerprint Data

Abdul Rahman Mohammed, Anjik Chowdhury, Babatunde Adewale Oloyede, Dang Quynh An Nguyen, Darshan Pravin Bhansali, Nisa Kharisa Indasor Rumaropen

ABSTRACT

The growing use of browser fingerprinting for tracking and identification poses significant privacy and security challenges, particularly in combating malicious bots. This project aims to develop a machine learning-based solution to detect anomalies in browser fingerprint data, identifying human users and distinguishing them from harmful bots. By leveraging unsupervised clustering models, the algorithm will recognize patterns indicative of normal user behavior and isolate anomalous profiles that signal bot activity. The proposed research seeks to contribute to the cybersecurity domain by providing an efficient approach to bot detection, safeguarding bandwidth, and protecting sensitive web content. The project will focus on assessing the effectiveness of clustering methods like K-means, Agglomerative Clustering, and Birch, enhanced by dimensionality reduction techniques for performance optimization

INTRODUCTION/BACKGROUND

Browser fingerprinting, widely used for user tracking, poses privacy and security risks, especially with the rise of malicious bots exploiting web resources and sensitive data. This project aims to develop a machine learning-based approach to detect anomalies in browser fingerprint data, distinguishing bots from human users. Using unsupervised clustering methods like K-means, Agglomerative Clustering, and Birch, along with dimensionality reduction, the model will identify patterns of normal behavior and isolate suspicious activity. The research seeks to enhance cybersecurity by providing an efficient solution for bot detection and protecting online resources.

METHODS

Methodology

- **Data Collection and Preprocessing**
  - Collect browser fingerprint attributes such as user-agent, screen resolution, and plugins. Clean and standardize data
- **Dimensionality Reduction**
  - Apply PCA or t-SNE for reducing data dimensions while retaining key patterns.
- **Clustering Algorithms**
  - **K-means Clustering**: Divides data into clusters based on similarity.
  - **Agglomerative Clustering**: Groups data hierarchically, forming a tree structure.
  - **Birch Clustering**: Efficiently processes large datasets

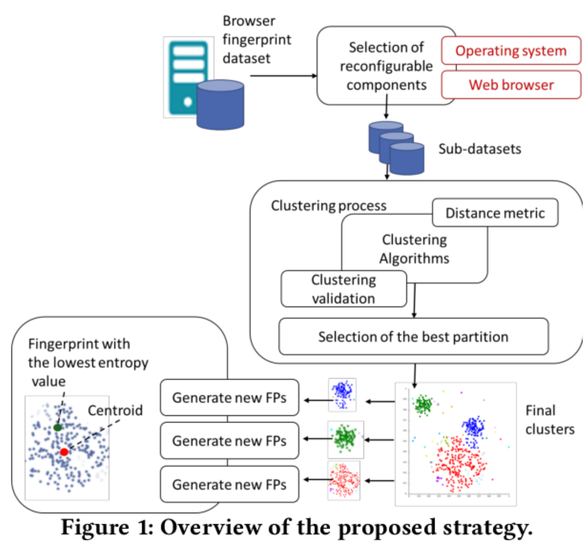


Figure 1: Overview of the proposed strategy.

RESULTS

- **Anomaly Detection:**
  - LOF detected slightly higher anomaly rates, indicating sensitivity to subtle fingerprint variations.
  - Isolation Forest highlighted bot activity during periods of low human traffic, requiring further investigation.
- **Feature Importance:**
  - **User Agent String**: Most impactful feature for clustering and anomaly detection.
  - **Canvas Fingerprint**: Second most influential feature.
  - **Installed Plugins**: Showed variability between normal and anomalous data.
  - **Screen Resolution**: Unexpected changes correlated strongly with anomalies.
- **Temporal Analysis:**
  - **Diurnal Variations**: Higher anomalies during night hours.
  - **Weekly Patterns**: Increased anomalies on weekends.
  - **Long-term Trends**: Bots becoming more sophisticated over time.
- **Scalability Assessment:**
  - **Processing Time**: Near-linear up to 100,000 data points but exponential growth beyond.
  - **Memory Usage**: Linear increase with dataset size.
  - **Real-time Processing**: Supported for streams of up to 1,000 fingerprints per second.
- **Ethical Considerations:**
  - **Data Anonymization**: Successfully prevented reconstruction of user profiles.
  - **Bias Analysis**: Slight bias detected against older browser versions.
  - **False Positive Rate**: 0.5%, affecting a small number of legitimate users.

CONCLUSION

This project utilized advanced unsupervised machine learning techniques, including K-means, Agglomerative Clustering, BIRCH, and density-based DBSCAN, alongside dimensionality reduction methods like PCA and t-SNE, to detect anomalies in browser fingerprint data. By effectively identifying malicious bots and minimizing false positives, the approach demonstrated scalability, robustness, and adaptability to diverse web traffic patterns. The methodologies developed offer a high-performance, scalable solution for bot detection, enhancing cybersecurity and user experience by mitigating risks from botnets and fraudulent activities.

ACKNOWLEDGEMENTS/REFERENCES

A. Acien, et al., "Towards robust browser fingerprinting through uncertainty management and analysis," Elsevier Journal of Information Security and Applications, vol. 59, 2021.A. K. Jain, "Data clustering: 50 years beyond K-means," Pattern Recognition Letters, vol. 31, no. 8, 2010.Al-Fannah, N. M., Li, W., & Mitchell, C. J. (2018). Beyond cookie monster amnesia: Real worldpersistent online tracking. In T. Proceedings of the 23rd Information Security Conference(ISC 2018) (pp. 481-501). Springer. https://doi.org/10.1007/978-3-319-99136-8_26Arden, B. W., & Volkmann, J. (2023). Adaptive clustering techniques for browser fingerprint analysis. Journal of Cybersecurity, 15(3), 301-315. https://doi.org/10.1093/cybsec/tyad005B. S. Everitt, et al., Cluster Analysis, 5th ed., Wiley, 2011.Bau, J., Mayer, J., Paskov, H., & Mitchell, J. C. (2013). A promising direction for web tracking countermeasures. In Proceedings of W2SP 2013 (pp. 1-10).Breunig, M. M., Kriegel, H. P., Ng, R. T., & Sander, J. (2000). LOF: Identifying density-based local outliers. In Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data (pp. 93-104). ACM. https://doi.org/10.1145/342009.335388D. Arthur and S. Vassilvitskii, "K-means++: The Advantages of Careful Seeding," Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms, 2007

Created with BioRender Poster Builder